Project 1: Energy Spent on Kylemore Abbey Week 1 Hikes

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Abstract

This report estimates the gravitational potential energy personally expended to summit four hikes completed during Week 1 at Kylemore Abbey: Famine Walk, Mamean, Croagh Patrick, and Sacred Heart Hike. Elevation changes and summit times recorded in the field are combined with body mass in a potential-energy model. MATLAB generates comparative plots of energy versus summit time and energy per hour. Results highlight that Croagh Patrick required the largest total energy and the highest sustained energy rate among the four hikes.

Contents

1 Introduction

The purpose of this report is to quantify, in an engineering sense, how much energy was required to overcome gravity while summitting four hikes completed during my time at Kylemore Abbey. Each hike presents different terrain, distance, and pacing conditions. By applying a consistent potential-energy model to documented elevation gains and summit times, this report compares hikes on equal footing and sets up a repeatable workflow, MATLAB for computation and plotting, Google Earth to gather information about the hikes and estimate the displacements covered during the hikes, and LATEX for presentation of all of the information I gathered.

2 Methods

2.1 Potential energy model

The approximate energy personally expended to summit each hike is modeled as gravitational potential energy gained from the net elevation change between the documented start and end points:

$$E = mg\Delta h \tag{1}$$

where (m) is body mass (kg), $g = 9.81\ \mathrm{m/s^2}$, and Δh is elevation gain (m).

This model intentionally neglects the energy lost gaining speed, horizontal locomotion, thermal regulation, wind, rest stops, elevation cancellation by going up and down the mountains, and route detours; however, those effects are discussed qualitatively in Section 4.

2.2 Field measurements

Elevation changes and summit times were recorded and compiled in ”Hike info.docx” and cross-checked against Google Earth KML exports where available (`nee sheep.kml` for Famine Walk, `Mamean.kml`, and `Croagh Patrick.kml`). Sacred Heart Hike is documented in the field notes only (but a KML export of it is available in the google earth listings, eventhough it is not referenced in the paper). Table 1 summarizes the values used in calculations. Body mass $m = 64.86$ kg (143 lb) was used for all plots.

Table 1: Elevation change and summit time for each hike

Hike	**Elevation change Δh (m)**	**Summit time**
Famine Walk	33.3	2 h 00 min
Mamean	171.3	45 min
Croagh Patrick	746.0	1 h 30 min
Sacred Heart Hike	144.5	45 min

2.3 Hike photographs

Field photographs from Week 1 hikes

Famine Walk coastal route near Kylemore (`Famine.HEIC`).

Mamean pass trail and summit approach (`Mamean.HEIC`).

Croagh Patrick pilgrimage trail on summit day (IMG_9652).

Sacred Heart Hike summit area near Kylemore (IMG_9688).

Figure 1: Week 1 hike photographs.

3 Results

Using Eq.1 with the Table 1 inputs, MATLAB computed the energies in Table 2. Figure 2 plots total potential energy versus summit time with distinct markers for each hike. Figure 3 compares energy per hour.

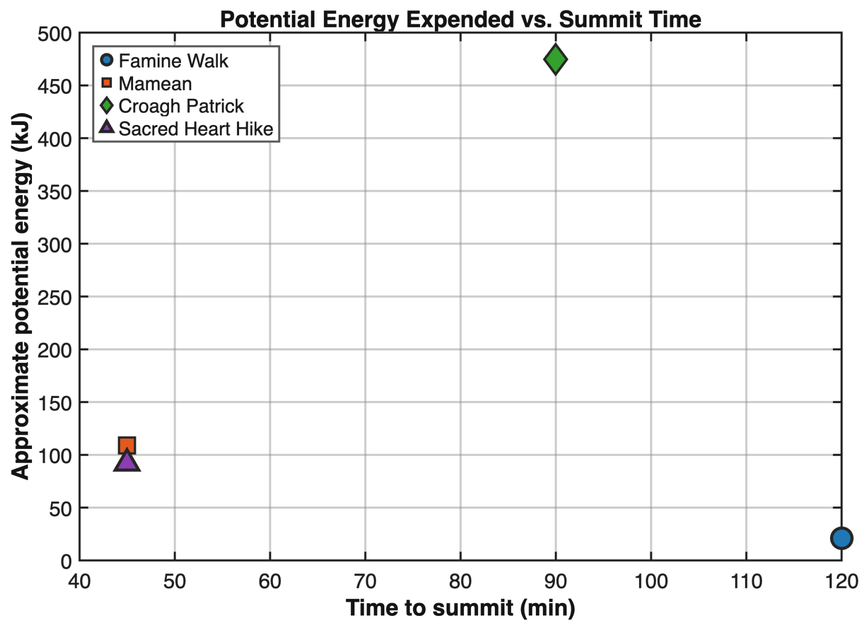


Figure 2: Approximate potential energy expended versus time to summit for each hike.

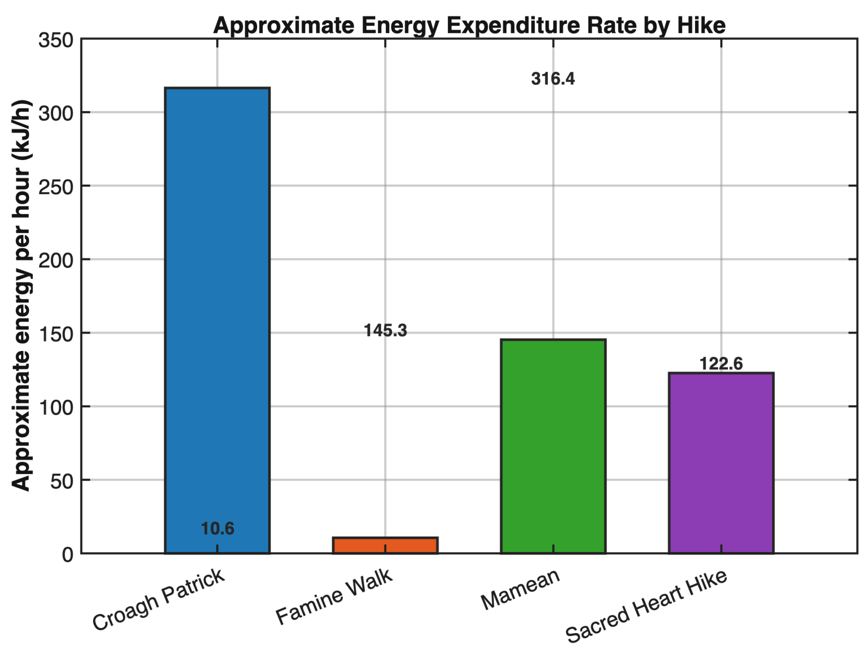


Figure 3: Approximate potential-energy expenditure rate (kJ per hour) for each hike.

Table 2: Computed potential energy and energy rate ($m = 64.86$ kg)

Hike	E_{pot} **(kJ)**	**Time (h)**	E_{pot}**/time (kJ/h)**
Famine Walk	21.2	2.00	10.6
Mamean	109.0	0.75	145.3
Croagh Patrick	474.7	1.50	316.4
Sacred Heart Hike	91.9	0.75	122.6

4 Discussion

Croagh Patrick dominates both total energy (475 kJ) and energy rate (316 kJ/h) because of its large documented elevation gain (746 m) combined with a relatively short summit window (1.5 h). Mamean ranks second in energy rate despite a smaller Δh than Sacred Heart Hike, since the same 45-minute summit time must deliver a larger vertical gain (171 m vs. 144 m).

Famine Walk has the lowest energy and lowest rate in this model because the net start-to-end elevation change (33 m) is small compared with the other hikes, even though the walk lasted two hours. Field notes also state that the route included unmodeled detours and elevation cancellations; the docx recommends a large margin of error for this hike.

Several limitations apply across all four hikes in this model. The most consistent source of error is the omission of variable hiking speed: on Mamean and Croagh Patrick in particular, a self-imposed pace meant higher speeds and therefore greater kinetic energy expenditure that the potential-energy model does not capture.

The model also relies solely on net elevation change from start to end, meaning intermediate downhill and uphill segments that cancel out are not accounted for this is especially significant on the Famine Walk, where the route involved numerous elevation fluctuations along the coastal path. A further unquantified event on the Famine Walk was an unplanned detour to climb a rocky region, for which no displacement or elevation data was recorded. Environmental conditions introduced additional error on Mamean and Croagh Patrick, where strong wind and rain required the body to expend energy on thermo regulation beyond what physical displacement alone would predict; this effect was more pronounced on Croagh Patrick given the steeper gradient and longer duration of exposure.

Finally, the Sacred Heart Hike's very short horizontal displacement (270 m) means small GPS or elevation measurement errors represent a proportionally larger fraction of the total, making that estimate less reliable. To partially offset these cumulative errors, margin-of-error adjustments were applied to each hike large for the Famine Walk and Croagh Patrick where uncertainty was highest, and smaller for Mamean and Sacred Heart where conditions were more controlled.

5 Conclusion

A consistent potential-energy model applied to four Kylemore hikes shows that elevation gain and summit time strongly control comparative energy metrics. Croagh Patrick required the most energy in this simplified analysis; Famine Walk required the least. The MATLAB scripts and LaTeX workflow used here—data in `hike_data.m`, reproducible figures, and structured reporting—will carry forward to later AME 34560 assignments that require documented analysis rather than hand-drawn plots.

References

[1] Elert, G. (2025), Potential energy, The Physics Hypertextbook, https://physics.info/energy-potential.

[2] AME34560, Lecture Notes, University of Notre Dame, 2026.

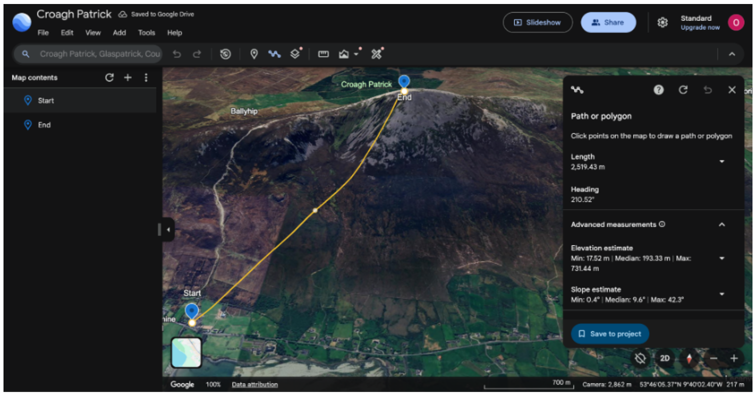
[3] Ineza, O., *Hike info.docx*, Field measurements, Kylemore Abbey Week 1, 2026.

[4] Google Earth, KML trail exports for Famine Walk, Mamean, and Croagh Patrick, 2026.

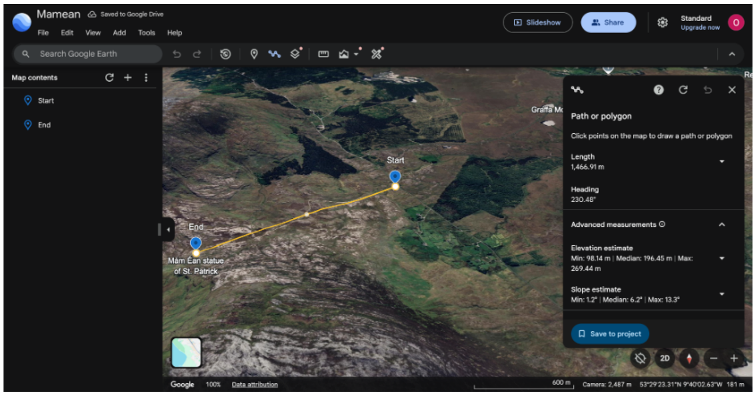
6 *APPENDICES*

6.1 Appendix A: (Google Earth Route Maps)

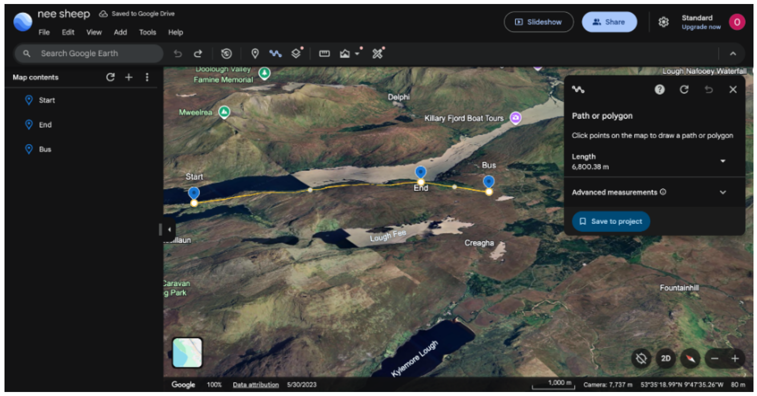
The following Google Earth screenshots document the way I recorded paths and elevation data for each of the four hikes. I consulted the local expert (Robert Black) in order to figure out where to drop the pins when finding the distances.



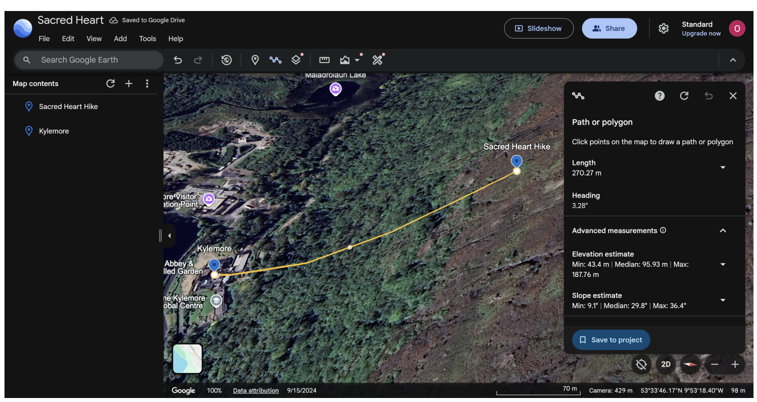
Croagh Patrick: Start to summit path. Length: 2,519.43 m. Elevation min: 17.52 m, max: 731.44 m. Median slope: 9.6.



Mamean: Start to end path. Length: 1,466.91 m. Elevation min: 98.14 m, max: 269.44 m. Median slope: 6.2.



Famine Walk: Start, end, and bus stop path along Killary Fjord. Total length: 6,800.38 m.



Sacred Heart Hike: Path near Kylemore Abbey. Length: 270.27 m. Elevation min: 43.4 m, max: 187.76 m. Median slope: 29.8.

6.2 Appendix B: [Hike info.docx] (field notes Written from DOCX File in Listings)

Famine Walk

Start: J4CR+52 Glassillaun, County Galway
Start Elevation: 3.2 m
End: H6W4+7P Leenaun, County Galway
End Elevation: 36.5 m
Displacement Start to End: 5,199.03 m
Displacement to Bus: 1,601.04 m
Bus place Elevation: 68.87 m
Time: 2 Hours
Limitations: The limitations to this model include the fact that I am not accounting for all the downhills I went through and the uphills I went through that were cancelled out. I also went astray to climb a rocky mountain and there is no way to track the energy used in this specific event. I also do not account for the different speeds I had when I was on the hike and it is inevitable that I used more energy speeding up.
Minimizing Error: To minimize error, I make a higher margin of error in the energy used in order to account for the energy I should have used. I use a very big value.
Link: `https://earth.google.com/web/@53.59702863,-9.87742209,256.39474323a,`
`16584.8472953d,35y,0h,0t,0r/data=`
`CgRCAggBMikKJwolCiExRjJFOFdxaEV4bUVRZUZBdndUS3hXcnFWbkVoeHhIMVcgAToDCgEwQgIIAEoHCOSTrF8Q`

Mamean Hike

Start: F8MJ+H4 Graffa More, County Galway
End: F8RW+WV Graffa More, County Galway
Displacement Start to End: 1,466.91 m
Displacement End to Start: 1,466.91 m
Elevation: 269.46 m − 98.16 m
Time: 45 minutes going up
Limitations: The limitations include the speeds I used in this hike. On this hike I challenged myself to move as fast as I could and this resulted in taking less time and using more speed, and hence more energy as I was on the hike. On this hike, it was so windy and this resulted into using more energy heating my body up. These factors carry less weight in the way they influence the energy used in the hike, so I made a model of minimizing error.
Minimizing Error: To minimize the Error, I make a small margin of error to help offset the amount of error made in the amount of energy I used.
Link: `https://earth.google.com/web/@53.47908334,-9.71800509,243.81629614a,`
`21396.10157981d,35y,0h,0t,0r/data=`
`CgRCAggBMikKJwolCiExSjRpMEdvaHBqNHV0WGh0Nk5hLWY0cXl6cFZLeXNZYXMgAToDCgEwQgIIAEoICLDk0-MH`

Croagh Patrick Hike

Start: Wild Atlantic Way Discovery Point @Croagh Patrick View, Murrisk Demesne, Co. Mayo
End: Q86R+28 Ballyhip, County Mayo
Displacement Start to End: 2,537.12 m
Displacement End to Start: 2,537.12 m
Elevation gained: 746 m
Time: 1 Hour 30 Mins
Limitations: The limitations include the speeds I used in this hike. On this hike I challenged myself to move as fast as I could and this resulted in taking less time and using more speed, and hence more energy as I was on the hike. On this hike, it was so windy and rainy and this resulted in using more energy heating my body up. These factors carried a lot of weight in the way they influence the energy used in the hike taking into consideration the fact this slope was very steep and it was, so I made a model of minimizing error.
Minimizing Error: To minimize the Error, I make a huge margin of error to help offset the amount of error made in the amount of energy I used.
Link: `https://earth.google.com/web/search/Croagh+Patrick,+Glaspatrick,+County+Mayo/@53.76977901,-9.64956792,193.35027338a,3780.46245439d,30y,0h,0t,0r/data=CiwiJgokCWNAM-6ZxOpAEfzdoeaJtOpAGZ-LsWdAASPAIcJeDhEdxCPAQgIIATIpCicKJQohMXV3RzNkUS1MUnZL authuser=0`

Sacred Heart Hike

Start: H466+M7 Shanaveg, County Galway
End: 53°33’50.62”N 9°53’20.67”W
Displacement Start to End: 270.27 m
Displacement End to Start: 270.27 m
Start Elevation: 43.4 m
End Elevation: 187.86 m
Time: 45 minutes going up

6.3 Appendix C: hike_summary.xlsx

Table 3: Hike energy summary — Kylemore Abbey Week 1 (`listings/hike_summary.xlsx`)

Hike	Elevation (m)	Time (h)	Energy (J)	Energy per hour (J/h)
Famine Walk	33.3	2.00	21,188	10,594
Mamean	171.3	0.75	108,994	145,326
Croagh Patrick	746.0	1.50	474,662	316,442
Sacred Heart Hike	144.5	0.75	91,917	122,555

6.4 Appendix D: MATLAB code for figures

hike_data.m

```
function data = hike_data()
    data.g = 9.81; data.mass_kg = 64.86;
    data.names = {'Famine Walk','Mamean','Croagh Patrick','Sacred Heart Hike'};
    data.dh = [33.3; 171.3; 746; 144.46];
    data.time_h = [2.0; 0.75; 1.5; 0.75];
end
```

project1_plots.m

```
%PROJECT1_PLOTS
close all; clc;
energy_kJ = (data.mass_kg * data.g * data.dh) / 1000;
eph_kJ    = energy_kJ ./ data.time_h;
time_min  = data.time_h * 60;

% Fig 1: Energy vs time
figure(1); hold on;
styles = {'or','sg','db','^m'};
for i = 1:4
    plot(time_min(i), energy_kJ(i), styles{i}, 'MarkerSize',10,'LineWidth',2,'DisplayName',data.names{i});
end
hold off; xlabel('Time to summit (min)'); ylabel('Potential energy (kJ)');
```

```
legend('Location','northwest'); grid on;

% Fig 2: Energy per hour
figure(2); bar(eph_kJ); set(gca,'XTickLabel',data.names);
ylabel('Energy per hour (kJ/h)'); grid on;
saveas(gcf, fullfile(fig_dir,'energy_per_hour.jpg'));

% Excel export
writecell([{'Hike energy summary - Kylemore Abbey Week 1'};
           {'Hike','Delta H (m)','Time (h)','Energy (J)','Energy/hr (J/h)'};
           [data.names', num2cell([data.dh, data.time_h, energy_kJ*1000, eph_kJ*1000])]], ...
           fullfile(appendix_dir,'hike_summary.xlsx'), 'Range','A1');
```